



PROJECT PROPOSAL

EE-312L POWER ELECTRONICS

PROJECT INSTRUCTOR

Muhammad Ahsan ul Haq

PROJECT TEAM

Rajed Mehmood [2023-EE-400]
Abdul Rafay Nadeem [2023-EE-410]
M Suleman [2023-EE-428]
M Meer Hamza [2023-EE-436]

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Project Proposal

Title: Design and Implementation of a Digitally Controlled DC-DC Buck Converter for Smart Energy Node Applications

1. Abstract

This project presents the design and implementation of a digitally controlled DC-DC buck converter for smart energy node applications. The system steps down a 12V DC input to a regulated and adjustable output voltage (5V–12V) using high-frequency PWM switching controlled by an ESP32 microcontroller. The converter incorporates real-time voltage and current sensing using an INA219 sensor module for monitoring power, efficiency, and protection.

The design focuses on hardware implementation, performance analysis under resistive loads, ripple evaluation, and efficiency measurement under continuous conduction mode (CCM) and discontinuous conduction mode (DCM). The proposed system serves as a foundational module for smart energy monitoring systems and future predictive fault detection applications.

2. Introduction

Power electronic converters are essential in modern electrical systems for efficient energy conversion and regulation. DC-DC converters are widely used in renewable energy systems, battery management, embedded electronics, and smart grids. Among various DC-DC topologies, the buck converter is commonly used for stepping down voltage levels efficiently using high-frequency switching devices.

Traditional analog-controlled converters lack flexibility in monitoring and protection. With the advancement of microcontrollers, digital control has enabled:

- Real-time monitoring
- Programmable duty cycle control
- Adaptive protection mechanisms
- Data logging for system analysis

This project aims to design a digitally controlled buck converter that integrates:

- PWM-based switching control
- Real-time current and voltage monitoring
- Efficiency analysis
- Fault detection capability

The system also acts as a prototype smart energy node aligned with future intelligent grid

monitoring applications.

3. Literature Review

DC-DC converters have been extensively studied for efficient power conversion. According to Erickson and Maksimović (*Fundamentals of Power Electronics*), buck converters operate by controlling the duty cycle of a switching device to regulate output voltage.

Studies show that:

- Increasing switching frequency reduces passive component size but increases switching losses.
- Low $R_{DS(on)}$ MOSFETs significantly improve efficiency.
- Schottky diodes reduce reverse recovery losses.
- Digital control improves flexibility and stability compared to analog control methods.

Recent research trends include:

- Microcontroller-based digital control of converters.
- Smart monitoring using current sensors such as INA219.
- Converter fault detection using real-time data analysis.
- Integration of DC nodes in microgrid architectures.

However, many laboratory implementations focus only on basic conversion without incorporating intelligent monitoring and protection. This project bridges that gap by combining power conversion and digital monitoring in a single hardware platform.

4. Proposed Methodology

The system consists of the following major stages:

4.1 Power Stage Design

A non-isolated buck converter topology is selected.

Design Specifications:

- **Input Voltage:** 12V DC
- **Output Voltage:** 5V–12V adjustable
- **Maximum Output Current:** 3A
- **Switching Frequency:** 25 kHz

Components:

- Logic-level MOSFET (IRLZ44N)
- Schottky diode (MBR1045)

- Inductor (150 μ H, 4A rated)
- Output capacitor (220 μ F + ceramic capacitors)

The ESP32 generates PWM signals to control the MOSFET switching.

4.2 Control Stage

The ESP32 microcontroller:

- Generates PWM at 25 kHz
- Adjusts the duty cycle to regulate output voltage
- Reads voltage and current via the INA219 sensor
- Calculates output power and efficiency
- Implements overcurrent protection

4.3 Sensing and Monitoring

The INA219 module measures:

- Load voltage
- Current
- Power

Measured data is:

- Displayed via Serial Monitor
- Logged for analysis
- Used for performance comparison

4.4 Protection Mechanisms

The system includes:

- 3A fuse (hardware protection)
- Software-based overcurrent cutoff
- Thermal consideration for MOSFET
- Proper decoupling capacitors

4.5 Experimental Procedure

1. Design and simulate the buck converter in MATLAB Simulink.
2. Calculate inductor and capacitor values.
3. Assemble hardware on a prototype board.
4. Test under resistive loads.
5. Record input/output waveforms.
6. Measure ripple voltage.
7. Calculate efficiency at different loads.
8. Compare simulation and hardware results.

5. Expected Results and Discussion

The expected output voltage relationship is given by the standard buck converter equation:

$$V_{out} = D \cdot V_{in}$$

Where D is the duty cycle.

Expected outcomes:

- Linear relation between duty cycle and output voltage.
- Stable operation in Continuous Conduction Mode (CCM) at high loads.
- Transition to Discontinuous Conduction Mode (DCM) at light loads.
- Output ripple within acceptable limits (<100 mV).
- Efficiency above 80% at moderate load.

The results will include:

- Duty cycle vs. Output voltage graph
- Efficiency vs. Load curve
- Ripple measurement
- CCM and DCM boundary analysis
- Thermal behavior discussion

Discussion will analyze:

- Switching losses
- Conduction losses
- Impact of inductor value
- Effect of switching frequency

6. Design Calculations and Future Work

6.1 Inductor Calculation

Calculated value:

$$L = \frac{V_{out} \cdot (V_{in} - V_{out})}{\Delta I_L \cdot f_s \cdot V_{in}}$$

Selected: 150 μH (standard value)

6.2 Capacitor Calculation

Calculated minimum:

$$C_{min} = \frac{\Delta I_L}{8 \cdot f_s \cdot \Delta V_{out}}$$

Calculated minimum: 45 μF

Selected: 220 μF (for low ripple)

6.3 Efficiency Estimation

Conduction loss:

$$P_{cond} = I_{out}^2 \cdot R_{DS(on)}$$

Switching loss:

$$P_{sw} = \frac{1}{2} V_{in} \cdot I_{out} \cdot (t_r + t_f) \cdot f_s$$

Total loss is used to estimate overall efficiency:

$$\eta = \frac{P_{out}}{P_{out} + P_{loss}}$$

6.4 Future Work

The system can be extended to:

- Bidirectional DC-DC converter
- Closed-loop PID voltage control

- Synchronous rectification
- IoT-based remote monitoring
- Fault prediction using AI models
- Integration into smart microgrid node (GridGuardian architecture)

This project forms the foundational hardware block for intelligent energy monitoring and predictive maintenance systems.